

React **useState** Hook

React **useState** Hook 允許我們追蹤 **function component** 中的狀態。
狀態通常是指應用程序中需要追蹤的數據或屬性。

元件程式最上方加入 **import the useState Hook**.

```
import { useState } from "react";
```

初始化 **useState**

useState 我們通過調用我們的函數組件來初始化我們的狀態。

useState 接受初始狀態並返回兩個值：

- 當前狀態。

- 透過函數更新狀態。

```
import { useState } from "react";
```

```
function FavoriteColor() {  
  const [color, setColor] = useState("");  
}
```

第一個值 **color** 是我們當前的狀態。

第二個值 **setColor** 是用於更新狀態的函數。

Read State

```
import { useState } from "react";  
import ReactDOM from "react-dom/client";
```

```
function FavoriteColor() {  
  const [color, setColor] = useState("red");  
  
  return <h1>My favorite color is {color}!</h1>  
}  
  
const root = ReactDOM.createRoot(document.getElementById('root'));  
root.render(<FavoriteColor />);
```

Update State

為了更新我們的狀態，我們使用我們的狀態更新函數

```
import { useState } from "react";
import ReactDOM from "react-dom/client";
```

```
function FavoriteColor() {
  const [color, setColor] = useState("red");

  return (
    <>
      <h1>My favorite color is {color}!</h1>
      <button
        type="button"
        onClick={() => setColor("blue")}
      >Blue</button>
    </>
  )
}
```

```
const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(<FavoriteColor />);
```

建立 FavoriteColor.js

```
import React, { useState } from "react";
import ReactDOM from "react-dom/client";
```

```
function FavoriteColor() {
  const [color, setColor] = useState("red");

  return (
    <div>
      <h1>My favorite color is {color}!</h1>
      <button
        type="button"
        onClick={() => setColor("blue")}
      >Blue</button>
      <button
```

```

        type="button"
        onClick={() => setColor("red")}
      >Red</button>
    <button
      type="button"
      onClick={() => setColor("pink")}
    >Pink</button>
    <button
      type="button"
      onClick={() => setColor("green")}
    >Green</button>
  </div>
);
}

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(<FavoriteColor />);
export default FavoriteColor;

```

修改 index.js

```

import React from 'react';
import ReactDOM from 'react-dom/client';
import './index.css';
import FavoriteColor from './FavoriteColor';
import reportWebVitals from './reportWebVitals';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(
  <React.StrictMode>
    <FavoriteColor />
  </React.StrictMode>
);
// If you want to start measuring performance in your app, pass a function
// to log results (for example: reportWebVitals(console.log))
// or send to an analytics endpoint. Learn more: https://bit.ly/CRA-vitals
reportWebVitals();

```

建立多個 state Hooks:

建立 Car.js

```
import { useState } from "react";
import ReactDOM from "react-dom/client";

function Car() {
  const [car, setCar] = useState({
    brand: "Ford",
    model: "Mustang",
    year: "1964",
    color: "red"
  });

  const updateColor = () => {
    setCar(previousState => {
      // ...previousState merge color , model into Car properties
      return { ...previousState, color: "blue" , model:"Porche 911" }
    });
  }

  return (
    <>
      <h1>My {car.brand}</h1>
      <p>
        It is a {car.color} {car.model} from {car.year}.
      </p>
      <button
        type="button"
        onClick={updateColor}
      >Blue</button>
    </>
  )
}

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(<Car />);

export default Car;
```

index.js

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import './index.css';
import Car from './Car';
import reportWebVitals from './reportWebVitals';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(
  <React.StrictMode>
    <Car/>
  </React.StrictMode>
);

// If you want to start measuring performance in your app, pass a function
// to log results (for example: reportWebVitals(console.log))
// or send to an analytics endpoint. Learn more: https://bit.ly/CRA-vitals
reportWebVitals();
```